Chapter 5

Implementation and Experimental Evaluation

This chapter addresses the fifth research objective: evaluating the complete interface in software-in-the-loop simulation and characterising where it fails. The evidence comes from five sources of different standing, which are kept apart throughout (Section 5.4.1). The primary results are those of a frozen, block-randomised cohort of 120 flights in the evaluated configuration of Section 5.2; two smaller frozen batches isolate the residual specific force and the estimator's inertia treatment; a second frozen campaign of 312 flights tests an independent force-balance release rule, uncommanded loss, a jammed gripper, thrust-map error and fixed and $\mathscr{L}_1$ baselines (Section 5.6); and development flights in earlier configurations are used only to show mechanisms. The frozen campaign of an earlier revision used a different estimator and eccentric grasps; none of its numbers is reused here.

5.1 Platform and Implementation

The implementation is organised as ROS 2 packages around PX4 software-in-the-loop [35] and Gazebo [36]. The controller node owns the reference generator, the predictive optimal control problem, the mission sequence and the release-confirmation logic; the estimator node owns the sliding measurement

window, the physical-input reconstruction, the mass optimisation, the presence machine and the published payload frame. The two communicate through the single atomic frame of Section 4.3.

Table 5.1: Main implementation components of the deployed pipeline.

Component	Role	Principal signals
Gazebo (Harmonic)	Rigid-body physics; detachable-joint gripper	pose, rotor speed, attach/detach record (logging only)
PX4 SITL	State estimation, rate loop, control allocation	odometry, actuator commands
`mhe_node`	14-state moving-horizon estimation; presence machine	odometry, rotor speed → payload frame (4.10)
`acados_nmpc_node`	Predictive control, mission sequence, confirmation	payload frame, state → thrust and body-rate setpoint
`payload_estimate`	Frame schema, evidence score (4.13)	shared by the online nodes and the offline replay
`proximity_gripper_node`	Grasp contact and release actuation	proximity, grasp enable, release command
`lifecycle_trace`	Machine-readable event trace for scoring	every state transition, with timestamps
`l1_adaptive`	Residual specific-force estimator (Section 4.5.3); also the comparison adaptive controller	odometry, rotor speed → $\hat{\boldsymbol{d}}$
`px4_pid_benchmark`	Bare-autopilot cascade baseline	position setpoints only

Every reported payload change is produced by the detachable-joint gripper, which adds or removes a genuine second rigid body and therefore changes mass and inertia together. A retired run-time inertia plugin did not alter the physics engine and contributes no result to this dissertation.

Across twenty recorded batches, reconstructed roll and pitch torques track the intended values (median correlations 0.93 and 0.70; slopes 1.08 and 1.10), whereas executed yaw torque is only 0.32 of the intended value. This three-fold yaw discrepancy is a concrete reason to drive the estimator with the measured wrench (3.24), not the command.

5.1.1 Platform parameters and task

The reusable-vehicle constants are

$$m_B = 2.0643\,\mathrm{kg}, \qquad J_B = \mathrm{diag}(0.0142, 0.0142, 0.0210)\ \ \mathrm{kg\,m^2}, \qquad k_d = 0.05, \tag{5.1}$$

with rotor arm $0.174\,\mathrm{m}$, $k_f = 8.55 \times 10^{-6}\,\mathrm{N\,s^2}$ and $k_m = 0.016\,\mathrm{m}$. The collective-thrust box is the actuator envelope (4.27), $[1.27\,\mathrm{N}, 31.35\,\mathrm{N}]$, and the torque bounds are $0.5\,\mathrm{N\,m}$ in roll and pitch and $0.2\,\mathrm{N\,m}$ in yaw. None of these grows with the estimate.

The value m_B appears in the model, in the derived quantity $m_P = (m_T - m_B)^+$ and in offline scoring. The payload is a $0.15\,\mathrm{m}$ cube of $0.15\,\mathrm{kg}$ or $0.30\,\mathrm{kg}$, placed directly below the grasp point so that the grasp is centred. The gripper attaches when the parcel is within $0.03\,\mathrm{m}$ horizontally of the attachment point and its vertical distance lies in $[0.15\,\mathrm{m}, 0.45\,\mathrm{m}]$. The vehicle descends to $0.35\,\mathrm{m}$ above ground to grasp, which realises a vertical arm of about $0.27\,\mathrm{m}$; both optimisers use the nominal prior $r_z = -0.27\,\mathrm{m}$, and the realised arm varies between about $0.25\,\mathrm{m}$ and $0.30\,\mathrm{m}$ from flight to flight (Section 3.4.4). None of the payload values is supplied to the estimator or the controller.

The task sequence is: approach under position control; hand-off to the predictive controller and warm-up hover; alignment over the parcel; controlled descent; grasp enable; lift from $0.35\,\mathrm{m}$ to $6.0\,\mathrm{m}$ in $8.4\,\mathrm{s}$; figure-eight carry (4.32) with $r = 10\,\mathrm{m}$, $z_h = 6.0\,\mathrm{m}$ and $d_z = 0.8\,\mathrm{m}$; release; recovery. The release command is issued after $45\,\mathrm{s}$ of loaded figure-eight, at the next tip of the figure-eight ($a = 3\pi/2$), so that every release occurs at the same geometric phase. The two speeds are set by the phase rate: $w = 0.283\,\mathrm{rad\,s^{-1}}$ for a peak of about $4\,\mathrm{m\,s^{-1}}$, and $w = 0.1415\,\mathrm{rad\,s^{-1}}$ with a $9.36\,\mathrm{s}$ speed ramp for about $2\,\mathrm{m\,s^{-1}}$. The scope of the controller is stated explicitly — it governs the grasp hover and the post-event adaptation phase, and the approach controller is not part of any claim. Batches run headless.

5.1.2 Real-time performance

On one desktop core shared with the simulator and the autopilot, the 13-state controller ($N = 20$, 20 Hz) solves in 1.3 ms at the median and 2.2 ms at the 99th percentile over $n = 1009$ solves: 2.6 % and 4.4 % of its 50 ms period. The controller's problem dimensions are unchanged by the estimator configuration.

The 14-state estimator ($N = 20$, 10 Hz) logs its solve time as a mean and a maximum over consecutive 2 s windows, not per solve. Over the 120 cohort flights, with the first two windows of each flight discarded ($n = 5800$ windows), the window mean had a median of 2.8 ms, a 90th percentile of 5.4 ms and a 99th percentile of 14.3 ms; the largest single solve recorded in any window was 77.3 ms. Against the 100 ms period the estimator never overran in the cohort, but the worst case used about three quarters of it, and because only window aggregates were recorded, the per-solve percentiles are not available.

The deadline is met in simulation. We nevertheless make *no* claim about headroom on an onboard computer; that requires measurement on the target hardware.

5.2 Evaluated Configuration

Throughout this dissertation, *evaluated configuration* means the 14-state mass-only MHE at 10 Hz ($N = 20$), driven by odometry and the rotor-speed wrench, with uniform stage weights, the $0.95\,m_B$ lower bound, lagged inertia (3.20), $r_z = -0.27$ m, window-balance seeding and the mass-only evidence score (4.13). It feeds the 13-state NMPC at 20 Hz. The cohort additionally sets five non-default choices explicitly:

1. the residual specific force $\hat{\boldsymbol{d}}$ of Section 4.5.3 ($a = 10$, $\omega_c = 0.5\,\mathrm{rad\,s^{-1}}$);

2. the attach-time inertia floor held until the release command (Section 4.6.3);
3. alignment relative to the parcel before grasp, to 0.012 m with at most 4 s of waiting (Section 5.8.2);
4. a smoothed initial yaw reference at the start of the figure-eight;
5. the shortened arm described above.

Offset compensation is off and every grasp is centred. Inner-loop scaling and lifecycle thresholds remain at their documented defaults; switches belonging to the unused moment/offset formulation are inactive. Appendix A gives the complete resolved configuration. All 120 flight records were checked and contain the 14-state selection, lagged inertia, residual, inertia-floor hold, relative alignment, yaw smoothing and stated r_z prior.

Result-version warning. The numerical cohort tables in this working revision were collected with the earlier product score, which multiplied the mass score by an inertia score derived from the same mass estimate. They remain useful as historical system data but do not constitute final closed-loop validation of (4.13). The cohort driver now selects the mass-only score explicitly; the tables and effect summaries in this chapter must be regenerated from that rerun before the dissertation is submitted. Offline replay found no timing change in 68 of 71 score-confirmed flights and a 0.1 s advance in the other three, but replay is not a substitute for the planned closed-loop rerun.

5.3 Experimental Questions and Hypotheses

The evaluation is organised around five questions. For each, the design of Chapters 3 and 4 implies an expected outcome, stated here so that the results

can be read against it.

Q1 Mass identifiability. Is the composite mass identified through grasp, carry and release from the measured wrench and odometry alone? *Expected:* accurate in loaded hover; biased low in sustained manoeuvre, because rotational model error reaches the mass through the attitude (Section 3.4.2).

Q2 Release completion. Is every physical release accepted, and how long after the payload has left? *Expected:* completion, at a delay of at least the 1 s persistence hold plus the time the estimate needs to fall.

Q3 Delayed detachment. When the command precedes the physical detachment, does an evidence-gated arm avoid accepting a release while the payload is still attached? *Expected:* the command-triggered arm accepts early by the length of the hold; the evidence-gated arms do not.

Q4 Arming policy. What does running the confirmation before the command cost in false acceptances during carry, and what does command-arming cost in delay? *Expected:* the always-on rule inherits the manoeuvre-induced collapse of the estimate (Section 4.4.6).

Q5 Tracking. Does the release arm change tracking, and what does the residual specific force contribute? *Expected:* no difference between arms, since the controller's mass follows the estimate in all of them; an improvement from the residual, which compensates the mass bias.

Failure modes and operating limits are examined separately in Section 5.8.

5.3.1 Release-acceptance arms

The release-acceptance logic is parameterised, so that the arms differ in exactly one factor wherever possible. Table 5.2 lists them. Arm B is the rule originally

proposed for this interface and the default of the implementation; arm C differs from it only in when the persistence count may start.

Table 5.2: Release-acceptance arms implemented as switchable settings. In every arm the controller's mass and inertia follow the estimate continuously; the arms differ only in when a release is accepted.

Arm	Name	Release-acceptance rule
B	Always-on confirmed	evidence score (4.14) counted throughout loaded flight; the command does not accept the release
A	Command-immediate	accepts the release at the command itself, with no confirmation
C	Command-armed confirmed	the same confirmation, but persistence counted only after the command arrives
—	Oracle	accepts at the true separation instant; an upper bound, not deployable, not run in the cohort

What *accepting* the release does should be stated precisely, because in the continuous interface it is less than the name of arm A suggests. Accepting marks the release complete in the lifecycle and resets the residual specific force $\hat{\boldsymbol{d}}$ of Section 4.5.3, which was compensating the loaded vehicle. It does *not* replace the controller's mass or inertia: in every arm those follow the published estimate throughout, so a vehicle whose payload is still attached keeps a loaded mass in its model whatever the arm has declared. The attach-time inertia floor is ended by the release command in all three arms (Section 4.6.3), after which the inertia, too, is whatever the estimate implies. The arms therefore differ only in *when* the release is declared complete and *when* $\hat{\boldsymbol{d}}$ is reset.

5.3.2 Conditions and scenarios

Three nominal conditions are used: 0.15 kg at $2\,\mathrm{m\,s^{-1}}$, 0.15 kg at $4\,\mathrm{m\,s^{-1}}$ and 0.30 kg at $4\,\mathrm{m\,s^{-1}}$, labelled c1, c2 and c3. The last is deliberately a stress condition, with hover at 74 % of maximum thrust, for reasons quantified in Section 5.8.3.

Two scenarios are crossed with those conditions in the frozen cohort:

1. **Nominal grasp–carry–release** (d0), arms A and B.
2. **Delayed detachment** (d4), arms A, B and C. A fault injector accepts the normal release command but holds the physical joint for 4 s. This separates "the command was issued" from "the payload left".

One further scenario was implemented but is outside the cohort. **Unsignaled loss**, in which the payload is detached at the release phase with no command and no notification, emulates a gripper that opens by itself or a load torn off. It was *not* run in the cohort, because in the evaluated configuration the attach-time inertia floor is released only by the release command (Section 4.6.3): after a loss without a command the empty airframe keeps flying with the envelope inner-loop scale, and smoke flights of this scenario showed pitch-moment saturation in 8–31 % of samples. A partial-loss scenario and a thrust-map mismatch scenario were also designed but were not run in the cohort; Section 5.9.2 states what is and is not known about them. Uncommanded loss and a thrust-map error in the release detector were later flown under the force-balance rule, which ends the inertia floor on detected loss (Section 5.6).

5.4 Cohort Design, Eligibility and Metrics

The cohort has five arm–scenario cells per condition, 15 cells in all. Each block runs every cell once, in an order drawn from a fixed random seed, and eight blocks give 120 flights. Every flight launched enters the statistics, and none is re-run: a flight that crashes or fails to grasp is reported as such, under the eligibility rule below. The arms are implemented by the release-baseline switch of the controller (A: accept at the command; B: always-on confirmation; C: command-armed confirmation). The cohort was launched on 30 October 2026 with block seed 1793358067; all 120 scheduled flights ran, and the attempts and exclusions are listed in Table 5.4.

Eligibility is declared before scoring: a flight enters the release statistics if the payload physically attached, according to the gripper's own attach record, and the vehicle survived to the release command. Physical attachment is used rather than the estimator's own mass evidence, because the latter would silently exclude flights whose estimator failed to recognise an attached payload — exactly the cases most worth reporting.

5.4.1 Sources of evidence

Table 5.3 lists every source of evidence used in this chapter. Only the first row supports the primary results of Section 5.5; the second, a frozen campaign of the same standing, is analysed on its own in Section 5.6 and never pooled with the first. The two smaller frozen batches each answer one question in one configuration; the horizontal-offset ablation was run in an earlier, eccentric configuration; and development flights are cited only for mechanisms, never pooled with frozen results or used to estimate an effect size.

Table 5.3: Sources of evidence in this chapter, in decreasing order of standing.

Source	Flights	Configuration and use	Section
Frozen cohort	120 attempted, 118 eligible	evaluated configuration; arms A/B/C; d0 and d4; three conditions; primary results	5.5
Release campaign	312 attempted, 312 eligible	evaluated configuration with the force-balance rule (arm D); four experiments, four conditions; analysed separately from the cohort	5.6
Residual ablation	48	arm B, nominal release, residual on/off, three conditions	5.7.1
Inertia-treatment comparison	16 (8 + 8)	0.30 kg at 4 $\mathrm{m\,s^{-1}}$; release at 45 s rather than at the tip	5.7.2
Horizontal-offset ablation	8 (4 + 4)	earlier eccentric configuration, 0.15 kg	5.7.3
Development flights	single flights to batches of 8	earlier configurations, changes confounded; mechanisms only	5.8.1, 5.7.2

5.4.2 Attempts and exclusions

Table 5.4: Attempts and exclusions in the frozen cohort. Eligible: physically attached according to the gripper's record, and survived to the release command. Crashed: before the release command. Conditions as in Section 5.3.2.

Cond.	Attempted	Attached	Eligible	Crashed	Excluded
c1	40	40	40	0	—
c2	40	39	39	0	1 missed grasp (A, d0)
c3	40	39	39	0	1 simulator stall (C, d4)
Total	120	118	118	0	2

Table 5.4 accounts for every flight launched. Two were ineligible. In one, the parcel was 0.47 m below the gripper at the grasp, just outside the attachment window — most plausibly the vertical drift of the grasp height described in Section 5.8.2 — and the grasp was never made. In the other, the simulator stopped advancing before take-off and the flight produced no estimator or controller data; this is an infrastructure failure, and it is reported rather than re-run, in accordance with the no-re-run rule of Section 5.4. No flight that produced data crashed or diverged in any phase.

5.4.3 Metrics

Position tracking is summarised by

$$\mathrm{RMSE}_p = \sqrt{\frac{1}{K}\sum_{k=1}^{K} \left\| \boldsymbol{p}_k - \boldsymbol{p}_k^{\mathrm{ref}} \right\|_2^2}. \tag{5.2}$$

Estimator metrics are the composite-mass error in loaded hover, in loaded manoeuvre and after release; the fraction of loaded figure-eight samples in which $\hat{m}_P$ falls below half the true payload, called the *collapse fraction*; the fraction of samples resting on a bound; and the number of window re-anchorings. The collapse threshold is relative because an absolute one does not transfer between payloads: a fixed 0.15 kg threshold, used in earlier development work at 0.30 kg

where it coincides with the relative definition, would classify every sample of a 0.15 kg flight with any negative bias as collapsed. Hover is the interval from 2 s after the end of the lift ramp to the start of the figure-eight; loaded figure-eight runs from 5 s after its start to the release command; post-release runs from 10 s to 40 s after physical detachment. Position RMSE in loaded figure-eight uses the same interval, and post-release RMSE the interval from 5 s to 35 s after detachment. Lifecycle metrics are the completion rate, the confirmation delay measured from the release command to the end of the 1.0 s evidence-score hold, and the count of confirmations occurring during loaded carry before any command or injected loss. Computational metrics are the median and upper percentiles of the solve time for both optimisers.

Ground truth is used only in offline scoring. The composite-mass truth is the airframe mass plus the payload mass while the simulator's joint record shows the payload attached; it never reaches the estimator.

5.5 Primary Results

5.5.1 Mass estimation

The behaviour the design intends is that $\hat{m}$ rises with no attach notification, holds near truth in loaded hover, and returns to the bare-airframe value after release without any event resetting it. The behaviour it does *not* guarantee is accuracy in sustained manoeuvre, where the estimator's approximate rotational model (Section 3.4.2) leaks into the mass. Figure 5.1 shows both on the nominal flights of the cohort, and Table 5.5 summarises every physically attached flight.

Three findings follow. First, in loaded hover the estimate is accurate in every flight: the per-flight mean error lies between $-0.44\,\%$ and $-0.10\,\%$ of the

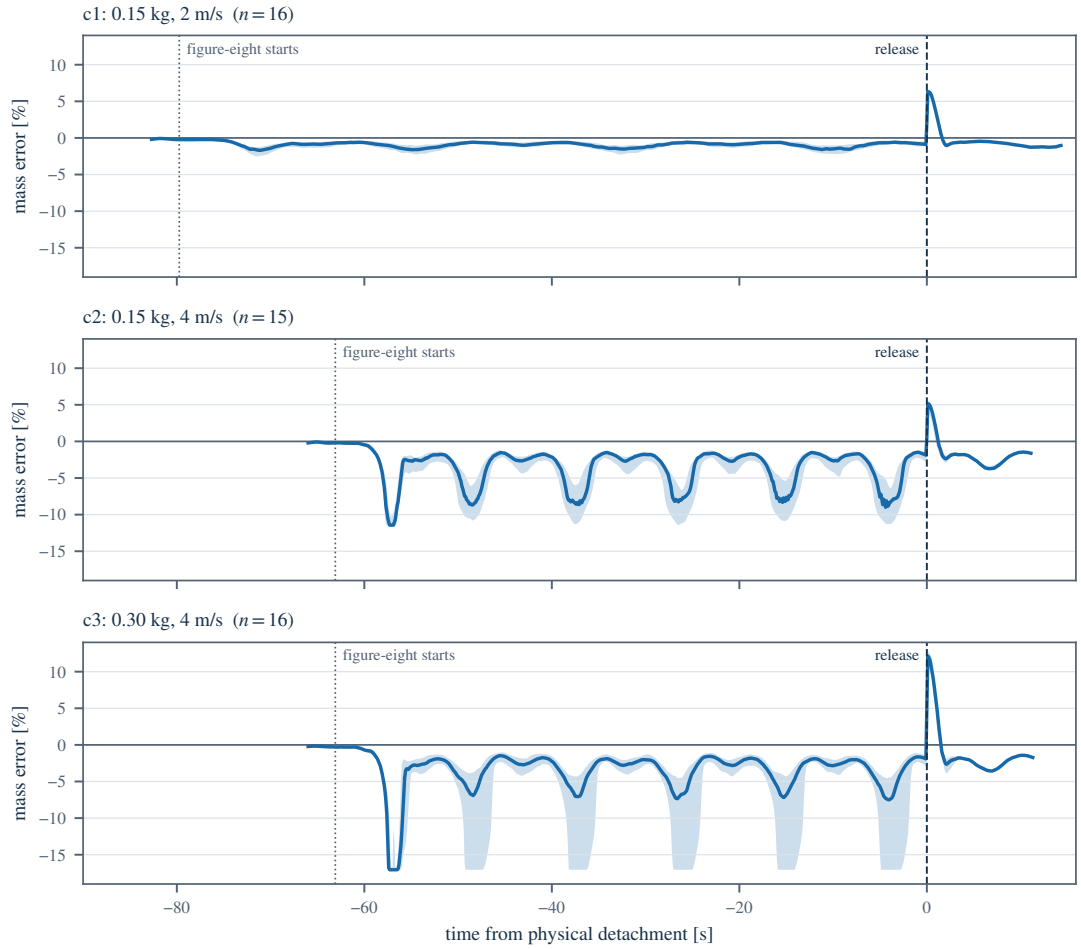


Figure 5.1: Composite-mass error $(\hat{m} - m)/m$ of the nominal (d0) cohort flights, each aligned on its own physical detachment and scored against its own truth. Line: median over flights; band: 10th–90th percentile. Traces start at the end of the lift ramp, because before lift-off the parcel rests on the ground and the joint record is not a usable truth. The positive transient at release is the estimator's lag behind an instantaneous truth step. The floor of the c3 band at $-17\,\%$ is the mass lower bound $0.95\,m_B$.

composite mass, which for the lighter payload is within about 10 g. No physically attached flight failed to recognise its payload. Second, in the figure-eight the bias is small at $2\,\mathrm{m\,s^{-1}}$ (median $-1.0\,\%$) and roughly four times larger at $4\,\mathrm{m\,s^{-1}}$, and it is not constant: the error dips about once every 11 s, half the figure-eight period, to $-6\,\%$ to $-9\,\%$ in the median trace. The first dip, at the entry to the figure-eight while the speed ramps up, takes the median of both $4\,\mathrm{m\,s^{-1}}$ conditions to the lower mass bound. Because the dip is a fixed fraction of the composite mass, it is a larger fraction of the *payload* for the lighter one, which is why the collapse fraction is highest at c2 (median 0.38)

Table 5.5: Composite-mass error by flight phase, all physically attached cohort flights (all arms and scenarios; the estimator is identical across arms). Errors are medians over flights of the per-flight mean, with the range of per-flight values in brackets; post-release error is relative to m_B. Collapse fraction: share of loaded figure-eight samples with $\hat{m}_P < \frac{1}{2}m_P$. Bound: share resting on $m_T = 0.95\,m_B$.

Cond.	n	Hover [%]	Figure-eight [%]	Post [%]	Collapse	Bound
c1	40	−0.21	−0.97 [−2.50, −0.47]	−0.94	0.00 [0, 0.12]	0.000
c2	39	−0.22	−4.09 [−6.27, −2.37]	−1.70	0.38 [0.21, 0.73]	0.010
c3	39	−0.29	−3.30 [−8.09, −1.62]	−1.68	0.03 [0, 0.40]	0.014

although c3's dips reach the bound more often. Third, after release the estimate settles slightly below the bare airframe, $-0.9\,\%$ to $-1.7\,\%$ of m_B; since $\hat{m}_P = (\hat{m} - m_B)^+$ is then zero, the release confirmation is unaffected.

The manoeuvre-induced dips of the second finding are the collapse that Section 5.8 discusses as the dominant failure of the interface. In the evaluated configuration the collapse fraction had medians of 0, 0.38 and 0.03 at c1, c2 and c3, with the per-flight maximum reaching 0.73 at c2 (Table 5.5); at $4\,\mathrm{m\,s^{-1}}$ the estimate touched the mass lower bound at the entry to the figure-eight in at least half the flights, and in some c3 flights at every dip thereafter (Figure 5.1). Its lifecycle consequence is reported in Section 5.5.4. The collapse no longer cost a flight: the inertia floor held the inner-loop scale at 2.34 through every loaded figure-eight (Section 5.8.3), the residual kept the vertical tracking error below 1 mm on average (Section 5.7.1), and no eligible flight diverged. But it was not prevented, and its effect on the lifecycle is undiminished. Development flights that show the mechanism of the collapse, in earlier configurations, are reported separately in Section 5.8.1.

5.5.2 Release completion and delay

Two delay modes are possible by construction and would have to be reported separately rather than pooled: confirmation while the vehicle is still flying the figure-eight, and confirmation after the 12 s timeout has declared UNRESOLVED

and the vehicle has braked to hover. In the cohort only the first occurred. Table 5.6 lists completion and delay for every eligible flight.

Table 5.6: Release completion and confirmation delay, all eligible cohort flights. Delay is from the release command to acceptance; under delayed detachment (d4) it is also given from the physical detachment. CP: two-sided 95 % Clopper–Pearson interval on the completion rate. Ranges are over flights.

Cond.	Arm	Scen.	Done	CP	From command [s]	From detachment [s]
c1	A	d0	8/8	[0.63, 1]	0	0
	A	d4	8/8	[0.63, 1]	0	−4.08
	B	d0	8/8	[0.63, 1]	2.96 (2.76–3.00)	2.96
	B	d4	8/8	[0.63, 1]	7.16 (7.10–7.26)	3.07
	C	d4	8/8	[0.63, 1]	7.13 (7.10–7.16)	3.07
c2	A	d0	7/7	[0.59, 1]	0	0
	A	d4	8/8	[0.63, 1]	0	−4.07
	B	d0	8/8	[0.63, 1]	2.60 (2.30–2.70)	2.60
	B	d4	8/8	[0.63, 1]	6.56 (6.30–6.76)	2.51
	C	d4	8/8	[0.63, 1]	6.65 (6.40–6.76)	2.62
c3	A	d0	8/8	[0.63, 1]	0	0
	A	d4	8/8	[0.63, 1]	0	−4.03
	B	d0	8/8	[0.63, 1]	3.03 (2.90–3.16)	3.03
	B	d4	8/8	[0.63, 1]	7.10 (6.96–7.26)	3.06
	C	d4	7/7	[0.59, 1]	7.00 (6.96–7.20)	2.97

Every one of the 118 eligible flights completed its release (CP interval $[0.969, 1]$); none reached UNRESOLVED, and no flight diverged. Pooled over conditions and scenarios, arm B completed 48 of 48 ($[0.926, 1]$). Arm B's confirmation arrived 2.3 s to 3.2 s after the payload physically left, in manoeuvre, with medians of 3.0 s at $2\,\mathrm{m\,s^{-1}}$ and 2.5 s to 3.1 s at $4\,\mathrm{m\,s^{-1}}$. About 1 s of this is the persistence hold of (4.14); the rest is the time the estimate takes to fall through the evidence-score transition band (Figure 5.1). That delay is the price of confirmation under nominal conditions, where the command is true.

The completion rate does not by itself show that the confirmations were correct. Arm B also confirmed releases that had not happened: 114 times during loaded carry, all at $4\,\mathrm{m\,s^{-1}}$, before any command. These are reported with arm C in Section 5.5.4; arm C confirmed real detachments in the same 2.4 s to 3.2 s range (Table 5.6).

5.5.3 Delayed detachment

A command-triggered arm accepts the release at the command and therefore declares, for the whole of the hold, that a payload has gone which is still attached. As Section 5.3.1 explains, in the continuous interface this does not put an empty model on the loaded vehicle: the controller's mass and inertia keep following the estimate, which still sees the payload. What the early acceptance does change is the lifecycle's report, which is wrong for the duration of the hold, and the residual $\hat{\boldsymbol{d}}$, which is reset while the mass bias it was compensating is still present. An evidence-gated arm should accept in none of these flights.

That is what the cohort shows. Arm A accepted before the joint was removed in all 24 of its delayed-detachment flights, about 4.0 s early in each; arms B and C accepted early in none of their 47 (Table 5.6). Measured from the physical detachment, arm B's confirmation came 2.3 s to 3.2 s later (medians 2.5 s to 3.1 s), the same as under nominal release: holding the joint for 4 s neither produced a confirmation during the hold nor slowed the one after it. What this experiment shows is a difference in *lifecycle correctness*. It does not show a difference in tracking or survival, and none was expected with the hold this short and the mass model unaffected; a permanently jammed gripper is a separate test.

5.5.4 Command-armed versus always-on confirmation

A delayed-detachment result reflects *persistent confirmation*: arm C, which starts counting only after the command, also never accepts early. The difference between arms B and C lies elsewhere. Arm B's confirmation rule runs throughout loaded flight and can fire during carry if the mass estimate collapses (Section 4.4.6); arm C's cannot, because it is not armed until the command. The

comparison therefore measures what running the rule before the command costs in false confirmations.

Table 5.7: Release confirmations during loaded carry (after physical attachment, before the release command), all physically attached cohort flights. Arm A does not run the confirmation rule.

Cond.	B: flights	B: confirmations	C: flights	C: confirmations
c1	0 of 16	0	0 of 8	0
c2	14 of 16	78	0 of 8	0
c3	6 of 16	36	0 of 7	0

Table 5.7 gives the result. Arm C confirmed no release during loaded carry in any of its 23 flights, a per-flight rate whose 95 % upper bound is 0.15. Arm B confirmed 114 releases that had not happened, in 20 of its 32 flights at $4\,\mathrm{m\,s^{-1}}$ and in none of its 16 at $2\,\mathrm{m\,s^{-1}}$. They are not isolated events: an affected flight typically confirmed about six times, spaced about 11 s apart, each time at an evidence score above 0.99. The spacing is the half-period of the figure-eight and coincides with the dips of Figure 5.1: each dip carries $\hat{m}_P$ below the confirmation threshold for longer than the 1 s hold. At c2 the light payload makes this nearly certain; at c3 the same relative dip leaves more payload mass above the threshold, and it happened in fewer flights.

In the evaluated configuration these false confirmations had no effect on control. With no release command pending, an acceptance neither resets $\hat{d}$ nor ends the inertia floor, and the controller's mass follows the estimate in either case; the release command re-arms the rule, so the genuine release was still confirmed in every flight. The error is in the lifecycle's report: a delivery system that acted on *release confirmed* would, at $4\,\mathrm{m\,s^{-1}}$, have been told repeatedly that a parcel still on the hook had been delivered. Arm C avoids this by construction, at no cost in delay (Table 5.6); its price is that it cannot detect a release that was never commanded.

The comparison settles the arming policy. The always-on rule of arm B is a negative result: it shows that, in this configuration, mass evidence alone does

not separate a loaded vehicle in manoeuvre from an empty one, and a rule that admits only that evidence will accept releases that have not happened. The command-armed rule of arm C keeps the property the delayed-detachment experiment established — it does not accept a release while the payload is still attached — without the false acceptances, and without a longer delay after detachment. It is the mechanism the present evidence supports. Its scope is correspondingly narrower: it distinguishes a commanded release from a physical detachment, and it cannot by itself detect a load lost without a command.

5.5.5 Tracking performance

Under nominal conditions the command is true, so waiting for confirmation can gain nothing and costs a delay; arm A has zero confirmation delay by construction. The purpose of the nominal pairs is to establish that the always-on confirmed arm completes and tracks as well as the command-triggered one, not that it is better: in the continuous interface the controller's mass and inertia follow the estimate in both arms, so no tracking difference is expected.

Pairs are formed within a block at the same condition. Both arms completed every pair. Post-release position RMSE (median over pairs, arm A against arm B) was 0.028 m against 0.026 m at c1 ($n = 8$, Wilcoxon $p = 0.38$), 0.108 m against 0.088 m at c2 ($n = 7$, $p = 0.078$) and 0.104 m against 0.102 m at c3 ($n = 8$, $p = 0.74$). Loaded figure-eight RMSE, before any difference between the arms can act, was 0.029 m to 0.033 m at c1 and 0.118 m to 0.126 m at $4\,\mathrm{m\,s^{-1}}$ in both arms. No difference in post-release stability was detected; the c2 pair count is seven because one arm A flight missed its grasp (Table 5.4).

Tracking is therefore governed by the condition rather than by the release arm: loaded figure-eight RMSE is about 0.03 m at $2\,\mathrm{m\,s^{-1}}$ and about 0.12 m at $4\,\mathrm{m\,s^{-1}}$. What does change tracking is the residual specific force, whose contribution is isolated in Section 5.7.1.

5.6 Release Campaign with Force-Balance Evidence

The cohort leaves three questions open: whether release evidence that does not pass through the mass estimate removes the carry-phase false acceptances of the always-on rule; whether such a rule can detect a load lost without a command and remain safe when a commanded release fails; and how the interface compares with fixed-parameter and lumped-adaptive control. A second frozen campaign of 312 flights, run on 3–4 November 2026, addresses them with the force-balance rule of Section 4.4.5, here called arm D. Its results are reported separately from the cohort and are not pooled with it; the earlier cohort's logs were used only for offline replay before the campaign was frozen.

5.6.1 Design

Arm D is the dual-enabled force-balance rule: the score (4.18), consumed only when a release command is pending or the uncommanded-loss watchdog is armed. Arm A is the command-immediate arm of Table 5.2. The two arms share every other setting of the evaluated configuration of Section 5.2, and they differ only in the release lifecycle. A fourth condition, c4 (0.30 kg at $2\,\mathrm{m\,s^{-1}}$), completes a two-by-two mass–speed design with c1–c3 as defined in Section 5.3.2. Table 5.8 lists the four experiments.

In E1 each block contains one round of eight flights with nominal release and one with delayed detachment, both rounds covering the four conditions under both arms, with the order of the rounds and of the flights within them drawn from a recorded seed. In E2 the three controllers share the arm-D lifecycle, the inertia floor and the inner-loop scheduling, and differ only in their translational compensation. *Fixed* keeps the bare-airframe mass in the prediction model with no translational adaptation. $\mathscr{L}_1$ keeps the same mass and compensates with the residual $\hat{\boldsymbol{d}}$ alone. *MHE* is the evaluated configuration, with the estimated mass

Table 5.8: Experiments of the force-balance release campaign. All four conditions c1–c4 appear in every experiment. d0: nominal release; d4: detachment held 4 s **after the command.**

Experiment	Flights	Factors	Arm
E1 Release semantics	144	arm A/D $\times$ d0/d4 $\times$ 4 conditions; 9 paired blocks	A and D
E2 Controller comparison	72	Fixed / $\mathscr{L}_1$ / MHE $\times$ 4 conditions; 6 rounds; d0	D
E3 Thrust-map error	48	detector thrust scaled by 0.90 / 1.10 $\times$ 4 conditions; 6 rounds; d0	D
E4 Faults	48	uncommanded loss / jammed gripper $\times$ 4 conditions; 6 rounds	D

and $\hat{\boldsymbol{d}}$. In E3 the thrust (4.17) seen by the force-balance detector is scaled by a fixed factor; the plant, the controller's thrust map and the residual are unchanged, the payload mass and thresholds are not altered, and the factor actually applied is recorded in each flight's log (0.900 and 1.100 in 24 flights each). In E4 an *uncommanded loss* detaches the payload at the release phase with no command, no pending release and no notification to the estimator. A *jammed gripper* receives the normal command and never releases.

Thresholds, filters, arming and confirmation times, controller and trajectory parameters were frozen before the first flight. A fingerprint of the flight code was recorded at the start and checked before every flight; it did not change. A flight was eligible if the payload physically attached, the vehicle survived to the release command or the injected fault, the estimator, controller, gripper, inertial, rotor-speed and residual logs were complete, and the recorded configuration matched the schedule. False or missed confirmations, divergence, solver failures, large tracking errors, an UNRESOLVED jam and a failed recovery after loss count as results, never as exclusions; in E2 a divergence before the command counts as a controller failure. Ineligible flights would have been replaced, and only those; none occurred. All 312 scheduled flights attached and were eligible, so every cell has its planned size (nine per E1 cell, six per cell elsewhere). No flight in the campaign diverged. Post-release metrics here use a window from physical detachment (or the injected loss) to 25 s after it,

which differs from the cohort's window; post-release numbers are not comparable across the two.

5.6.2 Release semantics and carry-phase acceptance

Table 5.9: Release acceptance under the force-balance rule (E1). Median (range) over nine flights per cell. det.: time of acceptance relative to physical detachment (negative = before the payload left); cmd: relative to the release command. Every flight in every cell completed its release.

	A, d4	D, d0	D, d4	
Cond.	**det. [s]**	**det. [s]**	**cmd [s]**	**det. [s]**
c1	−4.05 (−4.09 to −4.04)	2.06 (2.00–2.15)	6.20 (6.10–6.26)	2.12 (2.05–2.21)
c2	−4.09 (−4.10 to −4.04)	1.94 (1.90–2.05)	6.10 (6.00–6.16)	2.01 (1.91–2.12)
c3	−4.09 (−4.09 to −4.00)	2.30 (2.20–2.44)	6.40 (6.25–6.45)	2.31 (2.20–2.44)
c4	−4.05 (−4.09 to −4.04)	2.40 (2.30–2.46)	6.46 (6.34–6.60)	2.41 (2.34–2.51)

Table 5.9 reproduces the delayed-detachment result of the cohort with the new evidence. Arm A accepted before the payload left in all 36 of its delayed-detachment flights, by 4.05 s at the median. Arm D accepted early in none of its 36 (95 % CP upper bound 0.10) and completed all 72 of its releases ($[0.950, 1]$). Its confirmation came 1.90 s to 2.51 s after physical detachment (median 2.20 s, 95th percentile 2.44 s). Holding the joint for 4 s did not change the delay measured from detachment. About 1 s of it is the persistence hold; the rest is the two low-pass stages of (4.17)–(4.18).

The delay depends on the mass and hardly at all on the speed. In an effect-coded two-by-two regression on the nominal flights, the heavier payload lengthened the delay from detachment by 0.35 s (± 0.02, $p < 10^{-15}$), the higher speed shortened it by 0.08 s ($p = 0.0006$), and the interaction was not detectable ($+0.02$, $p = 0.64$); under delayed detachment the effects were $+0.30$, -0.14 and $+0.01$ s ($p = 0.76$ for the interaction). A larger payload leaves a larger step for the filtered score to traverse before it crosses 0.90.

The comparison that matters for the arming policy is carry. In every arm-D

flight the watchdog armed after the lift and before the release command, a median of 3.2 s after the start of the lift (range 1.9 s to 3.9 s). From that point, the rule was able to accept a release at any time during carry, which is the always-on property that produced the cohort's 114 false acceptances. Across all 240 flights of the campaign flown with the arm-D lifecycle, the watchdog was armed in 228, and no release was accepted during loaded carry in any flight (95 % upper bound 0.016 per armed flight), including every armed flight at c2, the condition at which the mass evidence had failed in 14 of 16 cohort flights. The force balance therefore separates a loaded vehicle in manoeuvre from an empty one where the mass estimate did not. The twelve flights in which the watchdog did not arm are the thrust-map cases of Section 5.6.4.

Arm D's delay after detachment, 1.9 s to 2.5 s, is also shorter than the 2.3 s to 3.2 s of the cohort's mass-evidence arms. The two come from different batches and are not a paired comparison.

Tracking did not differ consistently between the arms. Of the eight within-block pairings of post-release RMSE (four conditions, two scenarios), seven showed no difference ($p \geq 0.07$, Wilcoxon, nine pairs each); at c3 with delayed detachment arm D's was higher by 0.014 m at the median ($p = 0.004$). As in the cohort, the result concerns the lifecycle, not tracking.

5.6.3 Uncommanded loss and a jammed gripper

Table 5.10 gives the fault results. The watchdog was armed before the loss in all 24 uncommanded-loss flights, and every loss was detected ($24/24$, $[0.86, 1]$), 1.8 s to 2.5 s after the payload left; none was declared before it. In every case the payload geometry was zeroed, the inner-loop scale returned from the floor value to that of the bare airframe, no planned release was issued afterwards, and no flight diverged; the largest post-loss position error had a median between 0.08 m and 0.20 m by condition. This is the scenario the cohort could not fly,

Table 5.10: Faults under the force-balance rule (E4), six flights per cell. Loss: payload detached with no command. Armed: watchdog armed before the loss. Delay: from detachment to the declared loss, median (range). Reset: geometry zeroed and inner-loop scale returned to the bare airframe. Jam: command issued, payload never released. False acc.: release accepted. Hover: UNRESOLVED, braked, then hovering below $0.5\,\mathrm{m\,s^{-1}}$.

	Uncommanded loss				Jammed gripper	
Cond.	**Armed**	**Detected**	**Delay [s]**	**Reset**	**False acc.**	**Hover**
c1	6/6	6/6	2.03 (1.76–2.10)	6/6	0/6	6/6
c2	6/6	6/6	1.94 (1.90–2.00)	6/6	0/6	6/6
c3	6/6	6/6	2.34 (2.20–2.50)	6/6	0/6	6/6
c4	6/6	6/6	2.38 (2.30–2.46)	6/6	0/6	6/6

because under the mass-evidence rules nothing other than the command ended the inertia floor (Section 5.3.2).

The jammed gripper is the converse test: the command is true and the detachment never comes. No release was accepted in any of the 24 flights ($0/24$, $[0, 0.14]$), and the payload model was never cleared. Every flight reached UNRESOLVED at the 12 s timeout, armed the braking reference and came to hover, with a largest speed in hover of $0.18\,\mathrm{m\,s^{-1}}$ over all flights.

5.6.4 Thrust-map error

Table 5.11: Thrust-map error in the force-balance detector (E3), six flights per cell. m_{acc} bias: median error relative to the true composite mass, in loaded carry and after release. γ_{acc} max: median over flights of the largest score in loaded carry. Delay from the command.

Gain	**Cond.**	**Bias, carry**	**Bias, post**	**γ_{acc} max**	**Armed**	**Done**	**Delay [s]**
0.90	c1	−10.4 %	−10.3 %	1.00	0/6	6/6	0.96
	c2	−10.7 %	−10.5 %	1.00	0/6	6/6	0.95
	c3	−10.7 %	−10.6 %	0.62	6/6	6/6	1.48
	c4	−10.4 %	−10.4 %	0.18	6/6	6/6	1.56
1.10	c1	+9.3 %	+9.8 %	0.00	6/6	0/6	—
	c2	+9.1 %	+9.8 %	0.00	6/6	0/6	—
	c3	+9.0 %	+9.8 %	0.00	6/6	0/6	—
	c4	+9.3 %	+9.8 %	0.00	6/6	0/6	—

The force-balance mass moved one-for-one with the injected error (Table 5.11), and the two directions failed differently, as the analysis of Section 5.9.1 predicts for the mass channel. A 10 % error in the composite mass is 0.22 kg to 0.24 kg, larger than the lighter payload and close to the heavier one.

*Overestimated thrust (*1.10*).* After release the empty vehicle read about 10 % heavier than it was, the score never rose, and none of the 24 releases was confirmed. Every flight reached UNRESOLVED, braked and hovered, without divergence. The failure is a miss, and it ends in the safe state the timeout was designed for. By the same arithmetic, an uncommanded loss would also go undetected; that was not flown.

*Underestimated thrust (*0.90*).* No release was accepted during carry and every release completed, faster than nominally. That outcome hides two failures. With the lighter payload the loaded vehicle already read as empty — the score was 1.00 throughout carry — so the watchdog never armed in any of the 12 flights: protection against uncommanded loss was silently lost. Confirmation followed the command by the length of the hold alone, because the score was already high; with delayed detachment, which E3 did not include, the same rule would be expected to accept before the payload left. With the heavier payload the watchdog armed, but the largest carry score reached 0.85 in one flight, close to the 0.90 threshold. The dual enabling prevented every false acceptance in E3; it cannot restore evidence the thrust map has removed.

5.6.5 Fixed, $\mathscr{L}_1$ and MHE controllers

All 72 flights completed their release without divergence or saturation of the collective thrust, with one solver failure in one flight, and the confirmation delay did not depend on the controller (Table 5.12). Flights were paired within rounds; with six pairs the smallest attainable two-sided Wilcoxon p is 0.031.

Table 5.12: Controller comparison under the force-balance rule (E2), nominal release. Medians over six flights per cell. Loaded RMSE: from 5 s into the figure-eight to the command. Post: detachment to 25 s after it. Delay from detachment.

Cond.	Controller	Loaded RMSE [m]	Post RMSE [m]	Post peak [m]	Delay [s]
c1	Fixed	0.087	0.031	0.090	2.00
	$\mathscr{L}_1$	0.032	0.030	0.077	2.03
	MHE	0.033	0.026	0.072	2.08
c2	Fixed	0.151	0.118	0.203	1.94
	$\mathscr{L}_1$	0.099	0.085	0.136	2.02
	MHE	0.123	0.102	0.163	1.97
c3	Fixed	0.218	0.130	0.211	2.25
	$\mathscr{L}_1$	0.114	0.107	0.198	2.29
	MHE	0.117	0.096	0.206	2.32
c4	Fixed	0.168	0.036	0.170	2.40
	$\mathscr{L}_1$	0.030	0.045	0.149	2.40
	MHE	0.028	0.033	0.134	2.40

Against the fixed-parameter controller the benefit of adaptation is clear in loaded carry: the MHE controller had the smaller loaded RMSE in all 24 pairs ($p = 0.031$ in each condition), by 0.04 m to 0.14 m at the median, largest with the heavier payload. After release, when the fixed model is again correct, the difference in RMSE shrank to 0.002 m to 0.033 m and was consistent only at c1 ($p = 0.031$).

Against the $\mathscr{L}_1$ controller, which compensates the same payload with the residual alone, no consistent difference in loaded tracking was found ($p \geq 0.16$ in every condition); at c2 the $\mathscr{L}_1$ controller was nominally the better in five of six pairs. After release the MHE controller had the smaller peak error at both $2\,\mathrm{m\,s^{-1}}$ conditions in all six pairs ($p = 0.031$), by 0.006 m to 0.017 m, and the smaller RMSE at c4 ($p = 0.031$); at $4\,\mathrm{m\,s^{-1}}$ there was no consistent difference. With the residual present in both, estimating the mass is therefore not shown to improve tracking in carry, and its measurable effect in this experiment is a small one in the transient after release at low speed. The three controllers also share the inertia floor and the inner-loop scheduling, so E2 does not test the rotational benefit of the payload model.

5.6.6 Scope of the campaign

The campaign has the limitations of the cohort — simulation only, one airframe and gripper geometry, centred grasps, no wind — and some of its own. The thrust-map error was applied to the detector's input alone; neither the estimator's mass channel nor the plant was perturbed, and E3 used only nominal release. Every zero-count result is bounded by the cell sizes: an upper bound of 0.10 for early acceptance and 0.14 for false acceptance under a jam. The interface code of the campaign was not committed; the record holds the base revision, a patch of the uncommitted changes and an archive of untracked files, as for the cohort (Section 5.9.4). The campaign's data and scoring scripts are in `release_paper_20261103_210022/`.

5.7 Ablations

5.7.1 The residual specific force

The residual $\hat{\boldsymbol{d}}$ of Section 4.5.3 is part of the evaluated configuration, and its contribution was isolated in a separate frozen batch of 48 flights run on 1 November 2026 (block seed 1793469773). Arm B of the cohort was flown with the residual on and off, everything else in the evaluated configuration unchanged; the flight records confirm that the residual switch is the only setting that differs between the two arms. Nominal release (d0) was crossed with the three conditions, eight blocks each, in an order drawn from the seed. The endpoints were fixed before the batch: primary, loaded figure-eight position RMSE and mean vertical error; secondary, post-release RMSE, completion, divergence, mass bias and confirmations during carry. All 48 flights attached, completed their release and did not diverge.

Table 5.13: Residual specific force on against off (arm B, nominal release). Medians over eight within-block pairs per condition; "better" counts the pairs in which the residual-on flight had the smaller value (for the vertical error, the smaller magnitude); p: Wilcoxon signed-rank. Vertical error is $z - z^{\mathrm{ref}}$ averaged over loaded figure-eight.

Metric	Cond.	On	Off	Better	p
Loaded figure-eight RMSE [m]	c1	0.033	0.047	8/8	0.008
	c2	0.116	0.148	7/8	0.016
	c3	0.106	0.145	8/8	0.008
Mean vertical error [mm]	c1	$+0.4$	-4.5	7/8	0.008
	c2	-0.8	-28.4	8/8	0.008
	c3	$+0.2$	-44.3	8/8	0.008
Post-release RMSE [m]	c1	0.023	0.044	8/8	0.008
	c2	0.089	0.132	7/8	0.016
	c3	0.073	0.115	8/8	0.008

Table 5.13 gives the result. Without the residual the vehicle flies the loaded figure-eight below its reference, by 4.5 mm at c1 and by 28 mm and 44 mm at the two $4\,\mathrm{m\,s^{-1}}$ conditions; with it the mean vertical error is under 1 mm at every condition. Loaded figure-eight RMSE falls by 22 % to 31 % and post-release RMSE by 33 % to 48 %, in 23 of the 24 pairs on each metric. The sag grows with the manoeuvre-induced mass bias of Section 5.5.1, which is the error the residual is there to absorb: an estimate that is low by a few percent asks for a thrust that is low by the same fraction.

Equally informative is what the residual did not change. The loaded figure-eight mass bias was the same in both arms (medians $-0.9\,\%$ against $-0.8\,\%$, $-3.6\,\%$ against $-3.1\,\%$, $-3.1\,\%$ against $-3.5\,\%$; all $p > 0.19$), and so were the confirmation delays (all $p > 0.45$) and the confirmations during carry, which occurred in both arms at c2. The residual compensates the *consequence* of the mass bias for tracking; it does not reduce the bias, and it does not touch the lifecycle. This also qualifies the inertia-treatment result of Section 5.7.2, whose tracking endpoint was measured with the residual on in both arms.

5.7.2 The inertia treatment inside the estimator

Table 3.1 lists three treatments of the inertia increment inside the estimator. The lagged treatment is the deployed default; the evidence for that choice is a closed-loop comparison against the omitted treatment, reported below together with the configuration it was run in. The evidence against the live treatment is from single development flights.

Live. With the mass-dependent inertia in the estimator's rotational model, one $4\,\mathrm{m\,s^{-1}}$ flight overestimated the loaded mass by 28 % to 64 % relative to the force balance and diverged; with the airframe inertia additionally corrected to the simulator's value, one $2\,\mathrm{m\,s^{-1}}$ flight overestimated it by 29 % in the figure-eight and by 39 % after release, so that confirmation never arrived. These two flights motivated the rejection of the live treatment and the argument of Section 3.4.2.

Lagged against omitted. Sixteen flights were run in alternating order, eight per treatment, at 0.30 kg on the $r = 10\,\mathrm{m}$, $w = 0.283\,\mathrm{rad\,s^{-1}}$ figure-eight at a hover height of 6 m (about $4\,\mathrm{m\,s^{-1}}$ peak speed). The primary endpoint, fixed before the batch, was loaded figure-eight position error; the mass bias was secondary.

- All sixteen flights completed the release; there were no failures in either arm.
- Mean loaded figure-eight position error was 0.113 m with the increment omitted and 0.121 m with it lagged (Mann–Whitney $p = 0.23$): no detectable difference on the primary endpoint, and not in the lagged treatment's favour.
- The online loaded figure-eight mass bias was -8.9 % omitted and -4.0 % lagged (Mann–Whitney $p = 0.001$). The share of those samples resting on the lower mass bound was 9.8 % and 4.6 % ($p = 0.46$, not significant).

- Cross-replay removes the flight-to-flight variation: every one of the sixteen recorded flights was re-estimated offline under both treatments from the identical recorded inputs, and the lagged treatment had the smaller loaded figure-eight bias in all sixteen (Wilcoxon signed-rank $p = 3 \times 10^{-5}$). Replay fidelity of the recorded estimator against its own logged output was of order 10^{-7}.

The lagged treatment therefore halves the loaded mass bias in manoeuvre without changing tracking, and it was adopted as the default on that basis. It does not remove the bias.

Configuration of this comparison. Both arms ran in the evaluated configuration of Section 5.2 — residual specific force, inertia floor held until the command, shortened arm, alignment relative to the parcel and smoothed yaw reference — at 0.30 kg and $4\,\mathrm{m\,s^{-1}}$. It differs from the cohort only in the release timing: the command was issued 45 s into the loaded figure-eight rather than at the next tip. Two consequences should be kept in view. The result is a property of that configuration; with the earlier 0.47 m arm the increment is three times larger and the comparison has not been run. And because both arms carried the residual $\hat{d}$, which compensates the translational effect of a mass bias (Section 4.5.3), the tracking endpoint is partly insensitive to the bias by construction; the null tracking result does not show that the bias is harmless.

5.7.3 The horizontal-offset decision

The decision to hold the controller's horizontal centre-of-mass offset at zero (Section 3.3) rests on a control-side ablation performed with eccentric grasps, before the offset was removed from the model. The cohort used 0.15 kg, a nominal 0.10 m grasp offset and the $r = 10\,\mathrm{m}$, $\omega = 0.283\,\mathrm{rad/s}$ figure-eight, with four time-paired trials per arm in alternating order. The ablated arm set the

controller's offset parameters to zero, removing both the predicted thrust-offset moment and its trim-cost reference; the estimator and every other part of the controller were identical in both arms. All eight attempts physically attached and completed, with no solver failures.

Mean loaded figure-eight position RMS was 0.1230 m with compensation and 0.1213 m without it (-1.4%); the mean per-flight peak was 0.2760 m versus 0.2705 m, and lift RMS 0.4032 m versus 0.4039 m. Even at a grasp offset three times the attachment tolerance of the present configuration, modelling the offset in the controller bought nothing measurable. This is consistent with an analysis of flight logs from an earlier eccentric-load configuration, which found that the autopilot's rate integrator, not the predictive controller's torque, carried about 99 % of the trim moment even when the controller modelled the offset explicitly. This small, single-condition cohort is not an equivalence demonstration, and it does not address offsets large enough to consume most of the torque authority; the present configuration avoids that regime by aligning before grasp. The frozen logs, configurations and scoring scripts are in `../data/ecc_ablation_20260924_202511/`.

5.7.4 A negative result on learned stage weights

The parameterised generalisation of the retired event-triggered schedule (Section 4.2.4) was optimised by cross-entropy search in the loop, on the earlier mass-only estimator, and the resulting schedule showed no measurable benefit over the hand-designed rule in a paired ablation. The objective was flat over a broad neighbourhood of that rule.

This negative result is substantively useful, and is reported rather than omitted. What produced the benefit in the predecessor system was *noticing that the model had changed and removing stale data*; the precise magnitude of the post-trigger weights is weakly determined by the data. A learned schedule should therefore

not be presented as a superior learned controller. It is also part of the reason the mainline retired event logic altogether.

5.8 Failure Modes and Operating Limits

5.8.1 Manoeuvre-induced mass collapse: development evidence

The dominant failure of the mass-only interface is that the loaded mass estimate collapses toward the bare airframe during the figure-eight. Its size in the evaluated configuration is measured on the frozen cohort in Section 5.5.1. The evidence below is of a lower standard and serves a different purpose: it shows the mechanism and the consequences of the collapse. It comes from development flights at 0.30 kg on a figure-eight of radius 5 m at $w = 0.283\,\mathrm{rad\,s^{-1}}$, predates both the final cohort and the lagged default — every flight below ran with the payload inertia omitted from the estimator — and mixes configuration changes noted where they occur. It is not pooled with the cohort.

Table 5.14 condenses the three development batches.

Table 5.14: Development evidence for the mass-collapse mechanism. Configuration changes are cumulative and the first two changes in the second row are confounded.

Configuration	Attempts	Completed	Main observation
Early baseline	8	2	two missed-grasp failures; four loaded flights lost roll control after $\hat{m}$ and the inner-loop scale collapsed
Relative alignment + yaw smoothing	8	7	survival improved ($p = 0.041$), but collapse persisted and one flight failed during carry
Shortened arm, with both changes above	8	8	collapse fraction fell from 0.05–0.90 to 0.03–0.24 ($p = 0.028$), but was not eliminated

Across the sixteen flights used for the arm comparison, collapse fraction rose

with measured arm length (Spearman $\rho = 0.66$, $p = 0.005$), implicating the omitted inertia without making it the sole cause. Separate high-speed development flights recorded seven and nine carry-phase acceptances; holding the inertia floor preserved control authority but did not restore the mass estimate or the rule's discriminating power. The frozen cohort measures the current frequency of these failures in Sections 5.5.1 and 5.5.4.

These observations bound what the mass-only interface can claim. Its release confirmation is sound only while the mass estimate is: where the estimate collapses under manoeuvre, a confirmation rule that admits no other evidence will confirm a release that has not happened. The remedies available within the present architecture are the lagged inertia treatment of Section 5.7.2, now the default, which halves the bias but does not remove it; the inertia-floor hold and the residual specific force, both in the evaluated configuration, which address two of its consequences (loss of inner-loop authority and loss of thrust feedforward) without touching its cause; and the command-armed arm C, which removes the false confirmation by not running the rule before the command.

5.8.2 Grasp alignment

The centred-grasp assumption of Section 3.3 depends on the vehicle actually being over the parcel when the grasp is enabled. In development flights the autopilot's local position estimate drifted 2 cm to 4 cm per flight relative to the simulator's world frame, while the controller believed its horizontal error to be under 2 cm; with the 0.03 m attachment tolerance, roughly one flight in four or five missed the grasp. Tuning the controller cannot fix an error in the position it is given. An alignment step relative to the parcel, emulating a downward-looking camera, aligns the vehicle on the parcel's relative position to within 0.012 m, waiting at most 4 s, before the grasp is enabled; it is part of the evaluated configuration. The same drift acts vertically on the grasp height, which

is why the realised arm varies from flight to flight (Section 3.6).

5.8.3 The operating limit

0.30 kg at $4\,\mathrm{m\,s^{-1}}$ is the stress condition. Hover requires 74 % of maximum thrust. At the shortened arm the inertia ratio (3.15) is 2.34 nominally and at most about 2.7 over the realised arm range, well inside the body-rate scaling cap of 5.0; with the earlier 0.47 m arm it was 5.07, above the cap, and the inner loop could not be given the authority the load implied. The arm was shortened for that reason, so what remains at this condition is a thrust and torque margin, not an inner-loop one. This is the quantitative form of the feasibility boundary predicted in Chapter 3.

The cohort confirms that the remaining margin was not exhausted. The controller logs the inner-loop scale and the share of torque commands at their bounds over consecutive 5 s windows. In loaded figure-eight the inner-loop scale had a median of 2.34 and its window minimum never fell below 2.34 in any physically attached flight: the inertia floor, held until the command, kept the inner loop at the envelope scale however far the estimate dipped. Roll torque was never at its bound; pitch was at its bound in at most 0.3 % of samples on average; yaw was at its bound in about 2.5 % of samples at $4\,\mathrm{m\,s^{-1}}$ and never at $2\,\mathrm{m\,s^{-1}}$. The controller's solver reported at most one failure in any flight. These are window-level diagnostics, not per-sample statistics, and they say nothing about margin under disturbances the cohort did not include.

5.8.4 Estimator health

In every physically attached cohort flight the published mass left the empty value and settled within 0.44 % of truth in loaded hover (Table 5.5). The one flight whose estimate stayed at the empty value is the missed grasp of Ta-

ble 5.4, where that was the correct answer: the parcel was 0.47 m below the gripper, outside the 0.45 m attachment window, and the 90th percentile of the estimated payload was 4 g. That flight is also the cohort's only instance of the bootstrap's known failure (Section 4.6.3): the empty airframe flew the $4\,\mathrm{m\,s^{-1}}$ figure-eight with the envelope inertia floor, at a 2.34-fold inner-loop scale, and did not diverge. Estimator health at the moment of the release command was not scored separately.

One structural gap is known independently of the cohort: the phase-timed release command is *not* gated on estimator health. A release issued while the estimate is frozen leaves a model that no longer matches the vehicle after detachment. Gating release on estimator health is an obvious remedy and is not implemented.

5.8.5 Configuration and default-safety

The mass-only configuration is selected by an environment switch whose default in both experiment entry points is the 14-state form, and the estimator reports at startup which lifecycle path is active. The estimator node's own library default is still the 16-state form with an augmented first moment, which this dissertation does not use: a node started directly, without an entry point, runs that form. Entry-point defaults and library defaults are therefore not interchangeable, and statements about defaults in this dissertation refer to the entry points. A configuration that does not match the one described here must be detectable from the run record alone, which is why every run records the resolved value of every behaviour-affecting switch (Appendix B).

The cohort driver sets the five non-default settings of Section 5.2 explicitly for every flight, so the cohort does not depend on the entry-point defaults, which for those settings remain off; the lagged inertia treatment is an entry-point default. A flight launched from an entry point without the cohort driver there-

fore does *not* run the evaluated configuration. The 14-state selection is both an entry-point default and set explicitly by the cohort driver, and each flight's record marks it as explicitly set.

5.9 Threats to Validity, Missing Baselines and Reproducibility

All system-level evidence is software-in-the-loop. The leading threats are:

- **Thrust-map fidelity.** The plant generates thrust with the same square law the estimator inverts, and rotor speeds are exact. Neither a mis-calibrated thrust coefficient (Section 5.9.1) nor speed noise, transport delay or induced inflow [34] was tested in the evaluated configuration.
- **Clean state estimates.** The odometry available in simulation is better than a real vehicle's, and the measurement weights (4.3) are set from representative rather than measured noise levels. The horizontal drift of the local position estimate (Section 5.8.2) is the one deviation from clean odometry that has already been seen to matter.
- **Rigid attachment and centred grasp.** Gripper compliance and pendulum modes are not modelled, and every reported grasp is within the attachment tolerance of the thrust axis. Larger offsets are outside the claim. The realised vertical arm differs from the nominal prior by up to about 10 % per flight and is not measured per flight.
- **Single gripper geometry.** Every cohort flight uses the shortened arm. The inertia ratio grows with the square of the arm, and the earlier 0.47 m geometry, at which the heavier payload exceeds the inner-loop scaling cap, is outside the claim.

- **Confounded disturbances.** A sustained vertical updraft or downdraft is intrinsically indistinguishable from a payload change on a collective-thrust-only residual, and the mass-only interface has no second channel to separate them. Wind is not evaluated.

5.9.1 Thrust-map calibration: an analytical sensitivity

The mass channel recovers m_T from an *absolute* thrust, so it inherits the calibration error of k_f directly — nothing in the formulation cancels it. The relation is analytical: a relative error ε in the estimator's thrust coefficient scales the reconstructed thrust by $1+\varepsilon$, and through (3.7) the identified mass by the same factor. At $\varepsilon = \pm 5\%$ and the 2.2 kg composite mass of the lighter condition this is ± 0.11 kg — three-quarters of the 0.15 kg payload, and between four and five times the 0.024 kg confirmation threshold. The arithmetic predicts two failures: a coefficient reading low may leave the payload unrecognised, and one reading high may leave a phantom payload after release, so that confirmation never arrives. The direction matters: the second degrades toward UNRESOLVED, which is safe, the first toward an unrecognised payload, which is not. For the estimator's mass these remain predictions; no mismatch was applied to the estimator. The force-balance detector of Section 4.4.5 has the same one-for-one sensitivity, and a $\pm 10\,\%$ error in its thrust was flown: both predicted failures occurred, a miss ending in UNRESOLVED on one side and, on the other, a lighter payload read as absent, which silently disabled the uncommanded-loss watchdog (Section 5.6.4).

5.9.2 Experiments designed but not run

Three experiments were designed for this dissertation but not run in the evaluated configuration. They are listed so that their absence is explicit; no result is

claimed for any of them.

- **Thrust-map mismatch in the estimator** (k_f scaled by 0.95 and 1.05 in the estimator only). Its expected effect is the analytical sensitivity of Section 5.9.1; only the release detector's thrust was perturbed in the campaign (Section 5.6.4).
- **Partial payload loss** (four 0.05 kg boxes, three removed one at a time without a signal, against a no-loss control arm). This is the case a binary attached/detached abstraction cannot represent at all, and the one where a continuous interface would be expected to differ in kind.
- **Window-seed comparison** (the window force balance (4.9) against the instantaneous seed (4.6) at $4\,\mathrm{m\,s^{-1}}$). The only evidence for the window seed is the development observation of Section 4.2.3.

Unsignaled loss, the fourth scenario outside the cohort, was flown in the release campaign under the force-balance rule (Section 5.6.3); it has not been flown under the mass-evidence rules, for the reason given in Section 5.3.2. The open experiments are taken up in Chapter 6.

5.9.3 Missing baselines

The release arms of Table 5.2 and arm D isolate release semantics. A fixed-parameter predictive controller and an $\mathscr{L}_1$-type controller were compared with the MHE controller in the release campaign (Section 5.6.5), under one lifecycle and with the inertia-floor and inner-loop scheduling shared by all three. Because the $\mathscr{L}_1$ arm uses the same residual law as the evaluated configuration, that comparison isolates the payload model rather than the presence of adaptation; it found no consistent tracking benefit in carry. The evaluation still does *not* include incremental nonlinear dynamic inversion [21], the full $\mathscr{L}_1$-adaptive

NMPC of [20] with its rotational channel, or a change-detection procedure such as CUSUM [32] applied to the same signals under a matched false-alarm constraint. The last is the largest remaining gap for the release question: the force-balance rule is compared with the mass-evidence rules and with command-triggered acceptance, not with an optimised detector.

5.9.4 Reproducibility

Reproducibility is handled by construction rather than by discipline. Every run records its code identity, the resolved value of every environment switch that affects behaviour, the problem dimensions, horizons, weights, bounds, event timestamps, random seed and solver statistics. Runs terminated by numerical failure remain in the denominator. The evidence-score function used online and the one used in offline replay are the same function, so a recorded flight can be re-scored without re-implementing the logic.

The code identity is recorded as a repository revision *plus* a frozen patch of uncommitted changes and an archive of untracked files, not as a single commit. The cohort was not run from a clean revision: the working tree contained uncommitted edits to this dissertation's sources, which do not affect flight code, and the autopilot source tree contained modifications belonging to an unrelated project. Each flight's record accordingly marks an exact-commit claim as prohibited. Every flight can nevertheless be rebuilt exactly from the recorded revision, patch and archive, and that is the claim made here; no cohort result should be cited as having been produced by a particular commit.

5.10 Chapter Summary

This chapter addressed the fifth research objective with a frozen, randomised cohort of 120 software-in-the-loop flights, two smaller frozen ablations and a second frozen campaign of 312 flights. The answers to the five questions are, in brief: the mass is accurate in loaded hover and collapses periodically in sustained manoeuvre at $4\,\mathrm{m\,s^{-1}}$ (Q1); every one of the 118 eligible releases was accepted, 2.3 s to 3.2 s after physical detachment (Q2); under delayed detachment, the evidence-gated arms never accepted a release while the payload was attached, whereas the command-triggered arm always did (Q3); the always-on confirmation accepted 114 releases that had not happened during high-speed carry, and the command-armed confirmation none, at no cost in delay (Q4); and the release arm did not change tracking, while the residual specific force improved it without changing the mass bias (Q5). In the campaign, a force-balance release rule enabled by the command or by an airborne-load watchdog accepted no release during carry in 240 flights, never accepted before a delayed detachment, detected 24 of 24 uncommanded losses and accepted none of 24 jammed releases; it is, however, sensitive to the thrust map in both directions, and a 10 % underestimate disabled the watchdog for the lighter payload. Adaptation clearly beat the fixed-parameter controller in carry, while estimating the mass gave no consistent tracking benefit over the residual alone. No estimator-side thrust-map, partial-loss or hardware result exists. Chapter 6 draws the conclusions these results support, and only those.